

What Are CBDCs and Why Do Central Banks Want Them?: QuickTake

By Bastian Benrath and Carolynn Look

(Bloomberg) -- Money is undergoing its biggest reinvention in centuries. With many consumers abandoning physical cash and cryptocurrencies evolving fast, central banks are on the move to ensure they don't fall behind on innovation. The goal is a digital form of legal tender that can compete with private-sector alternatives by being safer and cheaper to use. While many countries including the US haven't decided if they want to introduce a central bank digital currency, some are acting faster. China has rolled out its digital yuan already to 260 million users.

1. How do CBDCs work?

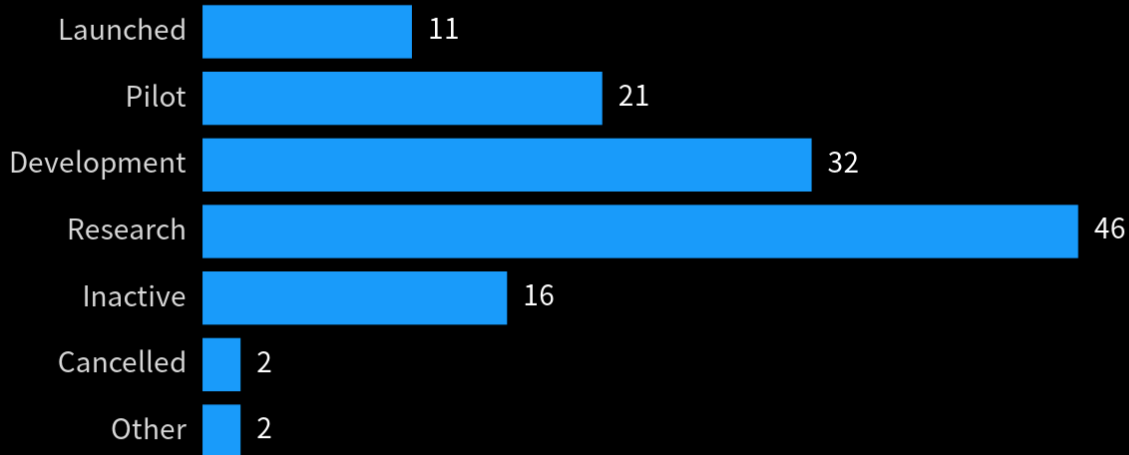
They're not so different, at least on the surface, from keeping money in a bank account and using plastic cards, smartphones or fintech apps to send it electronically into the world. The key difference to popular digital means of payment is that central-bank money – whether it's cash or a figure on a screen – is ostensibly risk-free while commercial bank deposits aren't. In theory, the money you hold in a bank account is convertible into paper cash on demand. In reality, that's subject to the health of the bank and its liquidity, meaning consumers might not always be able to access their balance and can lose money if a bank goes bust. CBDCs, like cash, would be a direct liability of the central bank, carrying its guarantee presumably for at least as long as the country behind it lasts.

2. What do CBDCs have to do with crypto?

Not much, apart from the blockchain technology powering crypto, which is also being used in some CBDC projects. Cryptocurrencies like Bitcoin are essentially a revolt against centralized, state-sanctioned finance, while CBDCs epitomize it. The central bank interest was initially seen as a response to private-sector pushes like then-Facebook Inc.'s 2019 announcement that it planned to introduce a global digital currency (first called Libra, later Diem, now dead). In some ways, the discussion has turned around, with issuers of cryptoassets now trying to capitalize on the credibility of central banks by developing so-called stablecoins – which try to maintain a steady price by pegging their value to an existing real-world currency. Still, given the volatility of unbacked crypto assets, the potential for CBDCs as a store of value and means of exchange is far greater. Crypto might play a role in Elon Musk's announced overhaul of X (née Twitter), but so far, the share of global payment transactions of even Bitcoin – the biggest cryptocurrency – is tiny.

Digital Currencies Picking Up

Number of countries and currency unions exploring CBDCs in June 2023



Source: Atlantic Council

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3. How would CBDCs work?

Central banks are experimenting on two main tracks: wholesale and retail. In the latter, consumers would have direct access to digital central-bank money, just as they nowadays hold cash. In wholesale projects, the focus is on a more-efficient technology for payments between banks and the central bank, possibly using blockchain. Retail CBDCs are particularly interesting for less-developed countries, where financial inclusion and even the logistics of distributing cash are a big issue. Most developed countries have instead focused on wholesale applications and are more reserved about retail because of their potential impact on commercial banking.

4. How would payments improve?

Digital payments could be settled cheaper and faster than they currently are – perhaps instantly, which would eliminate credit risk, and without the need for correspondent banks when crossing borders. In some countries where electronic payment methods are not very prevalent due to high costs for merchants, using a card or phone could become more widespread. That could foster competition among payment service providers so that even conventional credit card systems – like Visa or Mastercard – might become cheaper.

5. What are potential downsides?

In a retail CBDC setting, every citizen could theoretically have an electronic wallet at the central bank, eliminating the need for a regular bank account. That could undermine the broader financial system, since commercial banks are still needed for making loans and other important tasks. The European Central Bank has floated the idea of [capping the amount](#) of digital euros an individual could hold so that

people are still inclined to keep their commercial bank accounts. However, some countries, including Denmark and Switzerland, have ruled out the retail model. Instead, they and others, including the US, have wholesale CBDCs on the [top of their agenda](#), where they see the biggest potential for improvements with the least risks. Separately, privacy advocates worry about a loss of anonymity and the potential for government surveillance if cash payments disappear. Every digital payment – including those done today in online banking or payment apps – leaves a traceable record.

6. Who besides China is already trying this?

China has been testing the digital yuan, or [e-CNY](#), for years and, while still in a pilot phase, already has the largest user base – about 260 million users across 28 cities. Authorities there pushed for adoption of the digital currency after having declared all crypto-related transactions [illegal](#). Nigeria's eNaira went into circulation in late 2021 and has proven an [effective fall-back option](#) to paper money after a chaotic policy change resulted in a crippling [cash shortage](#). Nigeria is trying to avoid endangering banks by setting a low daily transaction limit, which is higher if users go through a commercial bank account. India's [digital rupee](#) pilot has some 1.6 million users as of mid-year and is planned to go wider. The Bahamas became a [global leader](#) in 2020 and Jamaica launched its [JAM-DEX](#) digital currency [last year](#). Among developed countries, [Switzerland](#) is set to test a real-world, wholesale CBDC for use between financial institutions for a limited period, starting sometime this year. That would put it farther down the road than its peers in the [euro area](#), the US, the UK and Japan, which are still in the exploratory phase.

Central Banks Go Digital

A sample of those using, developing or considering digital currencies

Central bank	Currency	Goals include	Status
People's Bank of China	e-CNY	Ensuring sovereignty while banning crypto	Pilot rolled out to 260 million users
European Central Bank	Digital euro	Ensuring sovereignty and complement cash	To decide this year whether to begin building
U.S. Federal Reserve	N/A	Understanding benefits and drawbacks of CBDCs	Conducting research
Central Bank of Nigeria	eNaira	Expanding basic financial services	Launched in October 2021
Central Bank of Brazil	Real Digital	Promoting investment and innovation	Pilot planned to start this year
Bank of Jamaica	JAM-DEX	Reducing cash handling costs	Launched in July 2022
Reserve Bank of India	e-rupee	Making transactions and payments more efficient	Pilot rolled out to 1.6 million users

Sources: Bloomberg, Atlantic Council

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7. Where is the US on CBDCs?

The [Federal Reserve](#) – long lukewarm on the idea – [said last year](#) that with a CBDC, the time it takes to settle foreign exchange transactions could be reduced from two days to under 10 seconds. A [white paper](#) published in May on a New York Fed project named Cedar disclosed that researchers have already conducted an experiment with a CBDC. Still, Fed Chair [Jerome Powell](#) made clear last year that for the US to enter the world of digital currencies, it would need executive and congressional approval and that this is “a process of at least a couple of years.”

The Reference Shelf

[Bloomberg QuickTakes](#) on [stablecoins](#), on [regulating crypto](#) and on [China’s digital yuan](#).

A [Bank for International Settlements survey](#) on where CBDCs are being investigated and a [proposal](#) on how digital central-bank money could work together with crypto. [Bloomberg Intelligence](#) surveys the [e-yuan landscape](#).

The Atlantic Council’s global [CBDC tracker](#).

[Bloomberg Opinion’s Lionel Laurent](#) on the push for a [digital euro](#).

A [blog post](#) of the [IMF](#) on how CBDCs fit into a future payment world.

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